

Zuhao Liu

PhD Student, Imperial College London

zuhao.liu25@imperial.ac.uk · zuhaoliu.github.io · [Google Scholar](#) · [GitHub](#)

Education

Imperial College London

Doctor of Philosophy (PhD) in Computing

London, United Kingdom

Dec 2025 – Present

Supervised by Tolga Birdal. Research focus: 3D vision, robotics, generative models, and world models.

Sun Yat-sen University

Master of Engineering in Computer Engineering

Guangzhou, China

Sep 2022 – Jun 2025

Advised by Prof. Wei-Shi Zheng. Research on video understanding, specifically in video anomaly detection.

University of Electronic Science and Technology of China

Bachelor of Engineering in Electrical and Electronics Engineering

Chengdu, China

Sep 2018 – Jun 2022

Advised by Prof. Guotai Wang and Dr. Huan Wang. Research on medical image analysis.

Work Experience

ByteDance – Machine Learning Engineer

Chengdu, China

Feb 2025 – Dec 2025

- Optimized user experience and developed models that significantly reduced customer complaint rates, impacting hundreds of millions of users.
- Skills: Recommender Systems, Large Language Models (LLM).

INSAIT – Graduate Student Researcher (Internship)

Sofia, Bulgaria

Aug 2024 – Jan 2025

- Conducted research on trajectory-controlled video generation for object interactions under the guidance of Prof. Luc Van Gool and Dr. Danda Pani Paudel.
- Proposed a novel video generation pipeline (InTraGen) interaction-based generation.

Baidu, Inc. – Autonomous Driving Engineer (Internship)

Beijing, China

Jun 2024 – Aug 2024

- Worked on autonomous vehicles and visual perception systems in the Intelligent Driving Group (IDG).

NetEase Games – Machine Learning Engineer (Internship)

Guangzhou, China

Dec 2022 – Nov 2023

- Led research on image-to-image translation and image generation; results published at ICCV 2023 as joint first author.
- Contributed to diffusion-based image generation models applied in game concept art production.
- Designed and implemented an interactive image cutout model deployed in game projects.

Publications

- [1] **Zuhao Liu**, M. Darcan, Z. Yu, R.A. Guler, T. Birdal. “HaNDF: Object-Conditioned Geometric Neural Hand Distance Fields.” *NeurIPS* 2026.
- [2] **Zuhao Liu***, A. Yanev*, A. Mahmood, I. Nikolov, S. Motamed, W.-S. Zheng, X. Wang, L. Van Gool, D.P. Paudel. “InTraGen: Trajectory-controlled Video Generation for Object Interactions.” *arXiv*, 2024. [\[arXiv\]](#) [\[Code\]](#)
- [3] **Zuhao Liu***, B. Cheng*, Y. Peng, Y. Lin. “General Image-to-Image Translation with One-Shot Image Guidance.” *ICCV*, 2023. [\[arXiv\]](#) [\[Code\]](#)
- [4] **Zuhao Liu**, X.-M. Wu, D. Zheng, K.-Y. Lin, W.-S. Zheng. “Generating Anomalies for Video Anomaly Detection with Prompt-Based Feature Mapping.” *CVPR*, 2023. [\[PDF\]](#)
- [5] X.-M. Wu, D. Zheng, **Zuhao Liu**, W.-S. Zheng. “Estimator Meets Equilibrium Perspective: A

Rectified Straight Through Estimator for Binary Neural Networks Training.” **ICCV**, 2023. [\[arXiv\]](#) [\[Code\]](#)

- [6] D. Zheng, X.-M. Wu, **Zuhao Liu**, J. Meng, W.-S. Zheng. “DiffuVolume: Diffusion Model for Volume Based Stereo Matching.” **IJCV**, 2024. [\[arXiv\]](#) [\[Code\]](#)
- [7] **Zuhao Liu***, H. Wang*, S. Zhang, G. Wang, J. Qi. “NAS-SCAM: Neural Architecture Search-Based Spatial and Channel Joint Attention Module for Nuclei Semantic Segmentation and Classification.” **MICCAI**, 2020. [\[Code\]](#)
- [8] **Zuhao Liu**, et al. “Automatic Attention Learning Using Neural Architecture Search for Detection of Cardiac Abnormality in 12-Lead ECG.” *IEEE Trans. Instrum. Meas.*, 2021.
- [9] **Zuhao Liu**, et al. “Automatic Segmentation of Organs-at-Risks of Nasopharynx Cancer and Lung Cancer by Cross-Layer Attention Fusion Network with TELD-Loss.” *Medical Physics*, 2021.
- [10] **Zuhao Liu**, et al. “CSAF-CNN: Cross-Layer Spatial Attention Map Fusion Network for Organ-at-Risk Segmentation in Head and Neck CT Images.” **ISBI**, 2020.
- [11] Y. Gao, H. Wang, **Zuhao Liu**. “An End-to-End Atrial Fibrillation Detection by a Novel Residual-Based Temporal Attention CNN with Exponential Nonlinearity Loss.” *Knowledge-Based Systems*, 2021.

Teaching Experience

Imperial College London – Graduate Teaching Assistant
Reinforcement Learning

London, United Kingdom
2026 – 2027

Sun Yat-sen University – Teaching Assistant
Linear Algebra and Statistics

Guangzhou, China
2023 – 2024

Honours and Awards

- **Tencent Scholarship**, Tencent 2023
- **First Prize**, Chinese Collegiate Computing Competition 2022
- **Outstanding Graduate**, UESTC 2022

Skills

Programming: Python, C, C++, Go, JavaScript, Flutter

Research: 3D Vision, Robotics, Video Understanding, Generative Models, World Models, Recommender Systems

Languages: Chinese (Native), English (Proficient)